CAMBRIDGE IGCSE CHEMISTRY 0620

Topic 3: Stoichiometry

Premium Answer Booklet

17 questions	2020-2025 Paper 4	Premium readable layout

Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

Contents

Question 01

7 marks

2021 May/June Paper 42 Question 3

Main topic: Topic 3: Stoichiometry
Answer source label: 0620_s21_ms_42.pdf

3(a)	• (thermal) decomposition
3(b)	• 84 (1) • 12.6 / 84 = 0.15(00) (1) • 0.15(00) / 2 = 0.075(00) (1) • 0.075(0) x 24.0 = 1.8 (1)
3(c)	• $Ca(OH)_2$ (1) • $CaCO_3$ (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 02 — 16 marks

2021 Oct/Nov Paper 43 Question 3

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 9: Metals, Topic 7: Acids, bases and salts, Topic 2: Atoms, elements and compounds, Topic 10: Chemistry of the
Answer source label: 0620_w21_ms_43.pdf

3(a)	• No separate point listed.
3(b)	• (207 / 239 x 100 =) 86.6%
3(c)	• PbS because the percentage of lead is larger in PbS or answer to (b) > 77.5%
3(d)	• PbCO3 -> PbO + CO2
3(e)	• PbCO3 + 2HNO3 -> Pb(NO3)2 + H2O + CO2
3(f)(i)	• two double bonds (1) • two pairs of non-bonding electrons on each oxygen and no non-bonding electrons on carbon (1)
3(f)(ii)	• bonds between ions or ionic bonds in lead(II) oxide (1) • attraction between molecules in carbon dioxide (1) • weaker attraction (between particles) in carbon dioxide opposite reasoning also allowed (1)
3(g)	• Mg + Pb2+ -> Mg2+ + Pb (1) • no reaction (1)
3(h)(i)	• glowing splint (1) • relights (1)
3(h)(ii)	• nitrogen dioxide / • nitrogen(IV) oxide

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 03 — 15 marks

2022 Feb/March Paper 42 Question 3

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 10: Chemistry of the environment, Topic 11: Organic chemistry
Answer source label: 0620_m22_ms_42.pdf

3(a)	• nitrogen (from air) and oxygen (from air) react • react due to high temperatures (of engine)
3(b)	• acid rain
3(c)(i)	• (thermal) decomposition
3(c)(ii)	• HCNO
3(c)(iii)	• (damp red) litmus • (litmus) turns blue
3(d)(i)	• 6NO2 8NH3 7N2 • either 6NO2 or 8NH3 • all three balanced
3(d)(ii)	• (nitrogen) loses oxygen
3(d)(iii)	• reduction and oxidation occur
3(e)	• Mr urea = 60 • 135 x 60 = 8100 and g to kg conversion = 8.1(00) kg
3(f)(i)	• carbon monoxide
3(f)(ii)	• carbon dioxide and nitrogen • February/March 2022

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 04 — 12 marks

2022 May/June Paper 41 Question 3

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 7: Acids, bases and salts, Topic 12: Experimental techniques and chemical analysis
Answer source label: 0620_s22_ms_41.pdf

3(a)(i)	• positive ions / cations (1) • sea of electrons / mobile electrons / delocalised electrons (1) • attraction between positive ions and electrons (1)
3(a)(ii)	• electrons move / • electrons mobile / • electrons flow
3(b)(i)	• ionic
3(b)(ii)	• ions move / • ions are mobile / • ions flow
3(c)(i)	• pipette
3(c)(ii)	• yellow to orange
3(c)(iii)	• at least two results are within 0.2 cm3 or less
3(c)(iv)	• 0.005 / 5 -> 10-3 (1) • 0.0025 / 2.5 -> 10-3 (1) • 0.125 (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 05

21 marks

2022 May/June Paper 42 Question 2

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_s22_ms_42.pdf

2(a)(i)	• Ca + 2H2O ■ Ca(OH)2 + H2 • H2 as product • Fully correct equation
2(a)(ii)	• calcium oxide
2(b)(i)	• 7 < pH ■ 12
2(b)(ii)	• OH-
2(c)(i)	• carbon dioxide
2(c)(ii)	• (a solution that) can dissolve no more solute • at a given temperature
2(c)(iii)	• add excess (solid) calcium hydroxide (to water) • filter
2(c)(iv)	• (aqueous) sodium hydroxide • white ppt • insoluble / remains in excess
2(d)(i)	• burette
2(d)(ii)	• neutralisation
2(d)(iii)	• indicator
2(d)(iv)	• Mol HCl = 0.0500 -> 20.0 / 1000 = 0.001(00) • Mol Ca(OH)2 = / 2 = 0.00100 / 2 = 0.0005(00) • -> 1000 / 25 = 0.0005(00) -> 40 = 0.02(00) • Mr Ca(OH)2 = 74 • -> = 74 -> 0.02 = 1.48 (g / dm3)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 06 — 14 marks

2022 Oct/Nov Paper 41 Question 4

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 12: Experimental techniques and chemical analysis, Topic 7: Acids, bases and salts
Answer source label: 0620_w22_ms_41.pdf

4(a)	• $CaCO_3 + 2HNO_3 \rightarrow Ca(NO_3)_2 + H_2O + CO_2$ • H_2O and CO_2 as product (1) • rest of equation correct (1)
4(b)	• fizzing / effervescence (1) • solid disappears / dissolves (1)
4(c)	• filtrate
4(d)(i)	• a solution that can contain no more solute (1) • at a given temperature (1)
4(d)(ii)	• cool the solution
4(e)(i)	• anhydrous
4(e)(ii)	• Mr $Ca(NO_3)_2$ = 164 (1) • mol $Ca(NO_3)_2$ = 2.46 / 164 = 0.015(00) (1) • 0.015(00) / 0.015(00) = 1 • 0.0600 / 0.015(00) = 4 and x = 4 (1)
4(f)	• $2NaNO_3 \rightarrow 2NaNO_2 + O_2$ • $NaNO_2$ on the right-hand side • equation completely correct

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 07

22 marks

2022 Oct/Nov Paper 42 Question 2

Main topic: Topic 3: Stoichiometry
Answer source label: 0620_w22_ms_42.pdf

2(a)(i)	• to prevent contact with air / oxygen and / or water
2(b)(i)	• combustion
2(b)(ii)	• yellow
2(b)(iii)	• 4Na + O2 -> 2Na2O • species (1) • balancing (1)
2(b)(iv)	• eight crosses in second shell of both Na (1) • six dots and two crosses in second shell of O (1) • ‘+' charge on each Na ion on correct answer line and ‘2-‘ charge on O ion on correct answer line (1)
2(c)(i)	• proton acceptor
2(c)(ii)	• pH 14
2(c)(iii)	• yellow
2(c)(iv)	• mol Na = 0.345 / 23 = 0.015(00) (1) • mol NaOH = = 0.015(00) (1) • -> 1000 / 50 = 0.015(00) -> 20 = 0.3(00) (1) • Mr NaOH = 40(1) • -> = 40 x 0.3= 12.(0) (g / dm3) (1)
2(d)(i)	• precipitation
2(d)(ii)	• red-brown
2(d)(iii)	• iron(III) hydroxide
2(d)(iv)	• Fe3+(aq) + 3OH-(aq) -> Fe(OH)3(s) • Fe(OH)3 (as only product) (1) • Fe3+ and 3OH- (as only reactants) (1) • state symbols (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 08 | 12 marks

2022 Oct/Nov Paper 42 Question 4

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 7: Acids, bases and salts, Topic 12: Experimental techniques and chemical analysis
Answer source label: 0620_w22_ms_42.pdf

4(a)	• $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$
4(b)	• no (more) fizzing (1) • solid stops dissolving or a solid remains / is visible (in the mixture) (1)
4(c)	• residue
4(d)(i)	• saturated
4(d)(ii)	• solubility (of $MgSO_4$ / solid) decreases (as temperature decreases)
4(e)(i)	• hydrated
4(e)(ii)	• Mr $MgSO_4$ = 120 (1) • mol $MgSO_4$ = 2.40 / = 2.40 / 120 = 0.02(00) (1) • 0.02(00) / 0.02(00) = 1 • 0.140 / 0.02(00) = 7 and x = 7 (1)
4(f)	• $2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$ • $NO_2 + O_2$ (1) • equation completely correct (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 09

12 marks

2023 Oct/Nov Paper 41 Question 4

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 6: Chemical reactions
Answer source label: 0620_w23_ms_41.pdf

4(a)	• test: relights • AND • observations: a glowing splint
4(b)(i)	• lower gradient (at t2)
4(b)(ii)	• concentration (of H2O2 particles) decreases (1) • frequency of collisions between particles decreases (1)
4(b)(iii)	• steeper curve which does not cross original curve and levels off before the original curve (1) • finishes at same volume (1)
4(c)	• mol O2 = 72 / 24000 = 0.003(00) (1) • mol H2O2 = -> 2 = 0.002 -> 2 = 0.006(00) (1) • conc H2O2 = -> 1000 / 20 = 0.006 -> 1000 / 20 = 0.3(00) (1) • Mr H2O2 = 34 (1) • = -> = 34 -> 0.3 = 10.2 (g / dm3) (1)
4(d)	• any transition metal oxide

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 10

17 marks

2024 May/June Paper 41 Question 5

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 4: Electrochemistry, Topic 7: Acids, bases and salts
Answer source label: 0620_s24_ms_41.pdf

5(a)(i)	• lead(II) nitrate(1) • sodium bromide / • potassium bromide / • ammonium bromide(1)
5(a)(ii)	• filter precipitate or lead(II) bromide or solid or residue(1) • wash the residue with distilled water AND dry e.g. between filter papers(1)
5(a)(iii)	• Pb2+(aq) + 2Br-(aq) -> PbBr2(s) • PbBr2 as the only product (1) • Pb2+ + 2Br- as the only reactants in a balanced equation(1) • state symbols • (aq) + (aq) -> (s)(1)
5(b)	• acidified aqueous potassium manganate(VII) (1) • purple to colourless (1)
5(c)	• 2CoO + 4NO2 (1) • 8H2O (1)
5(d)(i)	• heat again and weigh again • OR • repeat steps 2 and 3(1) • until mass is constant (1)
5(d)(ii)	• 0.005 / 5 ->10-3 (1) • 0.63 (1) • (0.63 / 18 =) 0.035 (1) • (0.035 / 0.005 =) 7 (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 11

15 marks

2024 May/June Paper 42 Question 3

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 5: Chemical energetics
Answer source label: 0620_s24_ms_42.pdf

3(a)	• Haber
3(b)	• enthalpy change (of reaction)
3(c)	• 450 and ■C • 200 and atm or 20 000 and kPa • iron
3(d)	• increase temperature • increase pressure
3(e)	• bond energy in breaking bonds • = [945 + (3 -> 435)] = (+) 2250 (kJ / mol) • = 2250 + 90 = 2340 • = 2340 / 6 = 390
3(f)(i)	• H2O
3(f)(ii)	• Ammonium chloride
3(f)(iii)	• mol CaO = 1.12 / 56 = 0.02(00) • mol NH3 = -> 2 = 0.02(00) -> 2 = 0.04(00) • vol NH3 = -> 24000 = 0.04 -> 24000 = 960

Easy explanation / reminder
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.
• Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.

END OF ANSWER

Question 12 | 12 marks

2024 Oct/Nov Paper 41 Question 4

Main topic: Topic 3: Stoichiometry
Answer source label: 0620_w24_ms_41.pdf

4(a)(i)	• +5 • +4
4(a)(ii)	• oxygen
4(a)(iii)	• Mr of Mg(NO3)2 = 148 • mol of Mg(NO3)2 = 7.40 / 148 = 0.05(00) (mol) • mol of NO2 = 0.05(00) -> 4 / 2 = 0.1(00) (mol) • volume of NO2 = 0.1(00) -> 24 000 = 2400 (cm3)
4(b)	• eight crosses in second shell of Mg • six dots and two crosses in second shell of O • ‘2+' charge on Mg ion on correct answer line AND ‘2-' charge on O ion on correct answer line
4(c)	• two dot-cross pairs as a double bond • four non-bonding electrons on each O

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.

END OF ANSWER

Question 13

14 marks

2024 Oct/Nov Paper 42 Question 4

Main topic: Topic 3: Stoichiometry
Answer source label: 0620_w24_ms_42.pdf

4(a)	• $Ag^{+}(aq) + Br^{-}(aq) \rightarrow AgBr(s)$
4(b)	• (silver) nitrate
4(c)	• 0.01 -> 1000 / 50 = 0.2(00) (mol / dm^3)
4(d)	• cream
4(e)	• mol of AgBr = 0.01(00) • Mr of AgBr = 188 • 188 x 0.01 = 1.88 (g)
4(f)	• sodium ethanoate
4(g)(i)	• rinsing of residue
4(g)(ii)	• (crystals of) sodium ethanoate
4(h)(i)	• named soluble barium salt e.g. barium chloride / barium nitrate • named soluble sulfate salt e.g. sodium sulfate / potassium sulfate / ammonium sulfate / sulfuric acid
4(h)(ii)	• any equation with $BaSO_4$ as a product • correct equation using soluble reactants

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 14

15 marks

2025 May/June Paper 41 Question 4

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 10: Chemistry of the environment, Topic 7: Acids, bases and salts, Topic 2: Atoms, elements and compounds
Answer source label: 0620_s25_ms_41.pdf

4(a)	• 3 dots and 3 crosses shared • one non-bonding pair of 2 dots on one N atom and 2 crosses on the other N atom to complete the octet on both
4(b)(i)	• Conditions can be in any order • (temperature) 450 and °C • (pressure) • 200 and atmospheres / 20 000 and kPa • iron and catalyst
4(b)(ii)	• $N_2 + 3H_2$ ■ $2NH_3$
4(c)(i)	• $5O_2$ -> $4NO + 6H_2O$
4(c)(ii)	• -3 • +2
4(c)(iii)	• increase in oxidation number
4(c)(iv)	• $4NO + 3O_2 + 2H_2O$ -> $4HNO_3$ • $HNO_3(1)$ • equation completely correct
4(d)	• 15 • 7.5 • 990

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 15

17 marks

2025 Oct/Nov Paper 41 Question 2

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 8: The Periodic Table
Answer source label: 0620_w25_ms_41.pdf

2(a)	• strontium / Sr
2(b)	• metallic • positive ions / cations • sea of electrons / mobile electrons / delocalised electrons • attraction between positive ions and electrons
2(c)(i)	• effervescence • solid dissolves • solution turns blue
2(c)(ii)	• calcium hydroxide (1) • hydrogen (1)
2(d)(i)	• orange-red
2(d)(ii)	• 2Ca + O2 -> 2CaO • CaO as only product • correct equation
2(e)(i)	• water(s) of crystallisation
2(e)(ii)	• 3.28 ■ 164 = 0.02(00) • 1.44 ■ 18 = 0.08(00) • =■=4

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 16

14 marks

2025 Oct/Nov Paper 41 Question 3

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_w25_ms_41.pdf

3(a)	• 2H2O • state symbols (aq) and (l)
3(b)	• neutralisation
3(c)	• 0.0100 -> 1000 ■ 40 = 0.25(0)
3(d)	• burette
3(e)	• mol H2SO4 needed = 0.0100 ■ 2 = 0.005(00) • vol of H2SO4 needed = 0.005(00) ■ 0.200 -> 1000 = 25 (cm3)
3(f)	• yellow • orange
3(g)	• repeat steps 1 and 3 without indicator
3(h)(i)	• (a solution containing the) maximum concentration of a solute dissolved in the solvent • at a specified temperature
3(h)(ii)	• solubility decreases as temperature decreases
3(h)(iii)	• none

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 17

13 marks

2025 Oct/Nov Paper 42 Question 3

Main topic: Topic 3: Stoichiometry
Cross-topic links: Topic 7: Acids, bases and salts, Topic 12: Experimental techniques and chemical analysis
Answer source label: 0620_w25_ms_42.pdf

3(a)	• -> $BaSO_4(s)$ + 2NaCl(aq) • 2NaCl (only) • (s) and (aq)
3(b)	• Mr $BaCl_2$ = 208 • 208 -> 0.1 = 20.8
3(c)	• 0.100 ■ 1.25 (-> 1000) = 80 (cm^3)
3(d)	• white
3(e)	• residue
3(f)	• rinsing the residue (with water) • (mass was greater) because (dry) residue contains solid sodium chloride / NaCl adds to the mass (of residue)
3(g)	• precipitation
3(h)	• named soluble barium salt other than barium chloride e.g. (barium) nitrate • named soluble sulfate salt other than sodium sulphate e.g. potassium (sulfate) / ammonium (sulfate) • named insoluble barium salt other than barium sulfate e.g. (barium) carbonate

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER